Hidden Currents: A Peak-Stress Water Index for Auditing AI Data Center Sustainability and Informing Mandatory Disclosure Policy

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***Abstract*—The water consumption of data centers has emerged as a critical sustainability concern, yet the metrics used to assess and report this consumption obscure two of its most important dimensions: temporal concentration (peak demand) and spatial heterogeneity (local water stress). The industry-standard Water Usage Effectiveness (WUE) metric is reported as an annualized fleet-wide average, treating every liter of water as ecologically equivalent. Recent research has separately quantified the peak-demand problem (Han et al. 2026) and the water-stress problem (Wu et al. 2025), but no existing metric integrates both. This paper introduces the *Peak-Stress Water Index* (PSWI), a composite metric that multiplies a facility's peak Water Usage Effectiveness (pWUE) by its location-specific Baseline Water Stress (BWS) value from the WRI Aqueduct 4.0 dataset. We apply PSWI to a new, publicly auditable dataset of 53 hyperscale and major colocation data centers operated by eight providers across North America, Europe, and Asia. Our findings reveal a 4,100× disparity between the best and worst facilities, demonstrate that company rankings shift by up to five positions when WUE is replaced by PSWI, and show that two facilities reporting identical annual WUE can differ in real water-stress impact by more than an order of magnitude. A Monte Carlo counterfactual simulation suggests that reallocating just 10% of fleet workload from worst-PSWI to best-PSWI facilities would yield a 33% reduction in fleet-wide water-stress impact. We conclude with a disclosure framework, modeled on SEC Regulation S-K, that would mandate quarterly per-facility PSWI reporting and propose enforcement mechanisms tied to existing public water system permitting authorities. All data and code are released under permissive licenses to enable independent verification and policy adoption.**

***Index Terms*—data centers, sustainable computing, water sustainability, environmental policy, water stress, AI infrastructure, disclosure**

I. Introduction

The rapid expansion of artificial intelligence has been accompanied by an equally rapid expansion of the data center infrastructure that powers it. Recent projections place U.S. data center electricity consumption at 6.7% to 12.0% of national totals by 2028 [1], and the International Energy Agency expects global data center electricity demand to grow at double-digit annual rates through 2030 [2]. Less widely discussed, but increasingly urgent, is the parallel growth in data center *water* demand. Between 2024 and 2030, U.S. hyperscale data centers alone are projected to require an additional 697 to 1,451 million gallons per day of water capacity—an amount comparable to the daily potable supply of New York City [3]. The total capital cost of expanding public water systems to meet this demand has been estimated at $10 to $58 billion [3].

These numbers, however, hide a deeper problem.

Industry's primary water sustainability metric, Water Usage Effectiveness (WUE) [4], treats every liter of consumed water as equivalent. WUE is reported as an *annualized* ratio of facility-level water consumption to IT energy, typically averaged over an entire year [5]–[7]. This treatment obscures two distinct dimensions of impact that are critical for assessing real-world water sustainability:

1) **Temporal concentration of demand (peak load).** Han et al. [3] demonstrate that data center water use is highly concentrated on the hottest days of the year, with daily peaking factors routinely reaching 6 to 10. A facility may report a low annual WUE while simultaneously placing extreme stress on local water systems during summer peaks, when demand from residential, agricultural, and industrial users is also at its highest.
2) **Spatial heterogeneity of stress (location).** The ecological and social cost of one liter of water consumed in Phoenix, Arizona—a region in the 22nd year of a megadrought affecting the lower Colorado River basin—is not equivalent to the cost of one liter consumed in Hamina, Finland, where sea-water cooling exploits effectively unlimited supply. Wu et al. [8] formalize this insight in their SCARF framework, weighting computing's water consumption by a location-specific stress factor derived from the World Resources Institute's Aqueduct dataset [9].

Critically, these two dimensions have been studied independently. Han et al. quantify peak demand but treat all water as equivalent. Wu et al. weight by stress but use annualized water totals. *No existing metric combines both.* Yet the policy implications of doing so are substantial: the worst-case water sustainability harm from a data center is its *peak* consumption multiplied by the *stress* at its location, a product that current annual, location-blind reporting renders invisible.

A. Contributions

This paper makes four contributions toward closing this gap.

a) (1) A composite metric.: We introduce the *Peak-Stress Water Index* (PSWI), defined as the product of peaking-adjusted Water Usage Effectiveness (pWUE) and location-specific Baseline Water Stress (BWS). PSWI integrates the two previously parallel research streams of Han et al. [3] and Wu et al. [8] into a single, comparable score (Section IV).

b) (2) An open, auditable global dataset.: We assemble the first publicly available cross-company dataset of hyperscale and major colocation data center water sustainability data, covering 53 facilities operated by eight major providers across North America, Europe, and Asia. All values are sourced from public sustainability reports, government records, and the open-source WRI Aqueduct 4.0 dataset; the full data and analysis code are released under permissive licenses (Section V).

c) (3) Empirical evidence of disclosure-driven misranking.: Applying PSWI to our dataset, we identify a systematic divergence between the rankings that companies present under self-reported WUE and the rankings that emerge under stress-weighted peak analysis. Several companies that appear "green" under WUE rank substantially worse under PSWI; the reverse is also true. We document a 4,100× disparity in PSWI score between the best- and worst-performing facility in our dataset, and we show that two facilities reporting the same annual WUE can differ in actual water-stress impact by more than an order of magnitude (Section VI).

d) (4) A policy framework for mandatory PSWI disclosure.: We propose a regulatory framework, modeled on the U.S. Securities and Exchange Commission's Regulation S-K climate-disclosure rules, that would require data center operators to publicly report PSWI on a per-facility, per-quarter basis. A Monte Carlo counterfactual simulation suggests that even a modest policy response—reallocating just 10% of fleet workload from worst-PSWI facilities to best-PSWI facilities—would yield a 33% reduction in fleet-wide water-stress impact (Section VII).

B. Organization

Section II reviews data center cooling, water metrology, and the regulatory landscape. Section III discusses prior work on geographical load balancing, carbon-aware computing, and water-stress-weighted accounting. Section IV defines the PSWI metric and analyzes its mathematical properties. Section V describes our dataset, sources, and methods. Section VI presents the empirical findings and case studies. Section VII develops the disclosure framework and counterfactual analysis. Section VIII discusses limitations and future work. Section IX concludes.

II. Background

A. Data Center Water Use: Why and Where

Data centers consume water both *directly* and *indirectly* [3], [4]. Direct (Scope 1) water use is dominated by on-site cooling. The high power density of modern GPU-based AI servers, often exceeding 30 kW per rack [1], produces heat loads that must be rejected to the outside environment. Two facility-level cooling architectures predominate: *evaporative cooling*, in which water is intentionally evaporated to remove heat at high thermodynamic efficiency, and *dry cooling*, which uses mechanical refrigeration or air-side economizers and consumes substantially more electricity for the same heat load. Industry disclosures indicate that water-cooled data centers can use 25–35% less electricity than air-cooled equivalents during peak summer load [3], [10]. Indirect (Scope 2) water use arises from upstream electricity generation, particularly thermoelectric power plants. While Scope 2 water typically exceeds Scope 1 in volume [4], it is rarely sourced from potable infrastructure and is therefore less directly relevant to the public water-system stress analyzed in this paper.

B. The WUE Metric and Its Limitations

Introduced by The Green Grid in 2011 and refined by Li et al. [4], Water Usage Effectiveness (WUE) is defined as:

$$\mathrm{WUE} = \frac{\text{On-site annual water consumption (L)}}{\text{Annual IT energy (kWh)}} \tag{1}$$

WUE is the metric universally adopted in corporate sustainability reports [6], [7], [10]–[12]. Its widespread adoption has enabled useful comparisons of efficiency across facilities and over time. But WUE has three structural limitations that compromise its policy relevance:

1) **Annualization erases peak signal.** Public water systems are sized to meet peak daily demand, not annual averages. Han et al. [3] document peaking factors of 6 to 10 for evaporative-cooled data centers in summer, and peaks above 10 in some planned facilities. A facility with WUE = 1.0 L/kWh and a peaking factor of 8 imposes a peak load equivalent to a WUE = 8.0 L/kWh facility on its host water utility—but the operator reports only the annual average.
2) **Location-blindness ignores ecological context.** A liter of evaporated water in Phoenix, Arizona—part of the Lower Colorado River basin which the U.S. Bureau of Reclamation currently classifies as in megadrought—is not ecologically equivalent to a liter of water in Hamina, Finland, where Google operates a sea-water-cooled facility on the Baltic [10]. WUE assigns equal weight to both, an accounting choice that masks orders-of-magnitude differences in real impact.
3) **Aggregation obscures facility-level harm.** Most operators report fleet-wide averages rather than per-facility values [6], [7], [11]. A company can dilute a single high-impact site by averaging it against many low-impact sites, yielding a favorable headline number while leaving the high-impact site's host community no recourse.

C. Public Water Systems and the Capacity Bottleneck

According to the U.S. EPA, public water systems in the United States face a 20-year infrastructure funding shortfall of $1.3–3.4 trillion [13]. Han et al. [3] project that meeting U.S. data center demand alone through 2030 will require

an additional 697–1,451 million gallons per day of new water capacity, valued at $10–58 billion—a substantial burden on systems that are already capacity-constrained, particularly in fast-growing data center hubs such as Loudoun County, Virginia and Maricopa County, Arizona. Building a new water source can take 20 years or more [3], far longer than the 3–5 year construction timeline of a hyperscale data center. Consequently, the limited *available* capacity of existing public water systems—rather than total volumetric demand—is emerging as the binding constraint on data center growth. This shifts the policy-relevant question from "How much water do data centers use?" to "How does data center water use stress specific public water systems at specific times?"—a question WUE is structurally unable to answer.

III. Related Work

Our work synthesizes three previously parallel research streams.

a) Sustainable AI and computing.: A growing literature quantifies the environmental impact of AI training and inference. Strubell et al. [14] were among the first to document the energy footprint of large language model training; Patterson et al. [15] extended this analysis to carbon emissions and showed that model size, hardware choice, and geographical location each contribute order-of-magnitude variation. Wu et al. [16] provided an industry perspective on sustainable AI from Meta's infrastructure team. Schwartz et al. [17] introduced the "Green AI" framing that explicitly weighs accuracy gains against compute cost. Luccioni et al. [18] performed the first detailed life-cycle carbon analysis of an open-source LLM (BLOOM). Most of this literature, however, treats water as either secondary to carbon or omits it entirely.

b) Geographical load balancing for sustainability.: A complementary line of work optimizes *where* computational workloads run to reduce environmental impact. Islam et al. [19] pioneered water-aware geographical load balancing for distributed data centers. Acun et al.'s Carbon Explorer [20] provides a holistic framework for designing carbon-aware datacenters. Radovanović et al. [21] describe Google's production carbon-aware computing system. Hanafy et al. [22] quantify the cost premium of carbon reduction in cloud workloads. Li et al. [23] introduce equity-aware geographical load balancing (eGLB), explicitly minimizing AI's environmental impact on the most disadvantaged region. Liu et al. [24] propose Geographical Server Relocation (GSR) as a complement to workload balancing. Bian et al. [25] apply carbon-aware federated learning across geographically distributed centers. Yang et al. [26] provide theoretical foundations for anytime-competitive constraint satisfaction in carbon-intelligent computing. Chien et al. [27] examine the carbon impact of generative inference. Across this literature, water is either absent or treated through unweighted volumetric metrics.

c) Water-aware computing.: The most directly related work is the recent emergence of water-stress-weighted accounting. Wu et al.'s SCARF framework [8] introduces the Adjusted Water Impact (AWI) metric, multiplying volumetric water consumption by a Water Stress Factor derived from WRI Aqueduct [9]. Jiang et al.'s WaterWise [28] co-optimizes carbon and water footprints in scheduling, while their ThirstyFLOPS work [29] provides region-specific HPC water modeling. Han et al. [3] provide the foundational quantitative analysis of data centers' impact on U.S. public water systems and propose the Peak Water Usage Effectiveness (pWUE) and Water Capacity Impact (WCI) metrics. The de Vries-Gao study [30] provides the most recent global estimate of AI carbon and water footprints (32.6–79.7 Mt CO_2 and 312.5–764.6 billion L respectively in 2025). Karimi et al. [31] examine water-energy tradeoffs in hot-arid climates. Siddik et al. [32] document the environmental footprint of U.S. data centers.

d) Our contribution.: The two structural deficiencies of WUE—peak blindness and location-blindness—have each been addressed in isolation: Han et al. [3] address peaks but treat all water as equivalent; Wu et al. [8] address location stress but use annual totals. *No prior work integrates the two.* We bridge this gap with PSWI and provide the first publicly auditable cross-company dataset to which the integrated metric can be applied.

IV. The PSWI Metric

A. Definition

We define the Peak-Stress Water Index for a data center facility i as:

$$\mathrm{PSWI}_i \;=\; \mathrm{pWUE}_i \;\cdot\; \mathrm{BWS}_i \tag{2}$$

where pWUE_i is the peak Water Usage Effectiveness as defined by Han et al. [3],

$$\mathrm{pWUE}_i \;=\; \mathrm{WUE}_i \cdot \beta_i, \tag{3}$$

with WUE_i the annualized water-usage effectiveness of facility i in liters per kWh, and $\beta_i \geq 1$ the daily peaking factor (the ratio of peak-day to average-day water consumption). The location-specific Baseline Water Stress BWS_i is the dimensionless ratio of total water demand to renewable supply at the facility's catchment, drawn from the WRI Aqueduct 4.0 baseline annual dataset [9].

PSWI thus has units of (L/kWh), with the interpretation: *the volume of water a facility consumes per kWh of IT energy on a peak day, weighted by the relative scarcity of that water at its location.*

B. Properties

a) Monotonicity.: PSWI is monotonically non-decreasing in each of its three input factors. A facility with worse WUE, a higher peaking factor, or a location with greater water stress will have a strictly higher PSWI, all else equal. This property ensures the metric responds to all three levers a regulator or operator might pull.

b) Comparability.: PSWI is dimensionally consistent across facilities and may be summed across a fleet (with workload weighting) to yield a fleet-level impact score, enabling cross-company benchmarking.

c) Decomposability.: For policy and reporting, PSWI decomposes naturally: $\log \text{PSWI} = \log \text{WUE} + \log \beta + \log \text{BWS}$. This separation allows operators and regulators to attribute changes in PSWI to efficiency improvements, peaking-factor reductions, or relocation decisions.

d) Edge cases.: For facilities in regions with BWS ≈ 0 (e.g., sea-water-cooled facilities in Hamina, Finland, where the relevant catchment is not freshwater-stressed), PSWI approaches zero, correctly indicating negligible freshwater-stress impact even if absolute water volumes are large. For facilities with BWS > 1 (extremely high stress, where demand exceeds renewable supply), PSWI amplifies WUE proportionally, correctly indicating that any consumption in such a location compounds an already-unsustainable resource depletion.

C. Interpretation thresholds

We propose the following PSWI interpretation thresholds, calibrated against our empirical distribution:

PSWI range	Interpretation
< 0.05	Negligible peak-stress impact
$0.05 - 0.5$	Low impact
$0.5 - 2.0$	Moderate impact (regulatory attention warranted)
$2.0 - 5.0$	High impact (mitigation required)
> 5.0	Critical impact (siting decision should be revisited)

These thresholds correspond approximately to deciles of the 53-facility distribution we present in Section VI. They are intended as starting values for regulatory deliberation, not as immutable boundaries.

V. Methodology

A. Dataset construction

We assembled a dataset of 53 hyperscale and major colocation data centers across eight operators: Google (16 facilities), Meta (9), Microsoft (8), Amazon (7), Equinix (6), Digital Realty (4), QTS (2), and xAI (1). Coverage spans the contiguous United States, Northern and Western Europe, and Asia (Singapore and Taiwan). All facility identifiers, locations, and reported WUE values are sourced from publicly accessible documents:

- Corporate sustainability reports for the 2024 reporting cycle [6], [7], [10]–[12], [33], [34].
- Government records, including U.S. county filings, water utility annual reports, and California Energy Commission docket records cited in Han et al. [3].
- The Lawrence Berkeley National Laboratory 2024 Data Center Energy Usage Report [1] for fleet-share weighting and IT-energy baselines.

For each facility we record: company, internal identifier, city, state/province, country, latitude, longitude, the most recent publicly-reported annual WUE in L/kWh, the source document, and the reporting year. The complete dataset is released as `datacenters.csv` in our public repository.

B. Water stress assignment

For each facility location, we assign a Baseline Water Stress (BWS) value drawn from the WRI Aqueduct 4.0 baseline annual dataset [9]. Aqueduct provides catchment-level BWS values as the ratio of total local water demand to total renewable supply, computed via the PCR-GLOBWB 2 hydrological model. We use the raw BWS ratio rather than Aqueduct's categorical labels (Low through Extremely High) to preserve quantitative resolution. For facilities in the United States, we use the published catchment values; for international facilities, we use the published country-level aggregates. The complete water-stress assignment table, with watershed identifiers and notes, is released as `water_stress.csv`.

C. Peaking factor estimation

Peaking factors are rarely disclosed by operators. Han et al. [3] document observed peaking factors of 6–10 for evaporative-cooled data centers in hot summer climates, with values exceeding 10 in some planned facilities. To remain conservative, we assign peaking factors based on Köppen climate zone:

Climate zone	Peaking factor β
Cool / subarctic	1.8
Mild oceanic	2.5
Temperate continental	4.0
Hot humid	6.0
Hot arid	8.0

Each climate-zone value is set at or below the lower end of Han et al.'s observed range for similar climates. This biases our PSWI estimates downward and is a deliberate methodological choice to ensure our findings are not driven by aggressive peaking-factor assumptions. A future revision of this work will use disclosed monthly water-use data where available (currently only Google publishes such data, and only at the fleet level [10]).

D. Reproducibility

All computations in this paper can be reproduced from the public repository at https://github.com/hrihaan19/pswi-data-center-audit. The pipeline consists of two Python scripts (`pswi_calculator.py` and `visualize.py`) depending only on `pandas`, `numpy`, and `matplotlib`. No proprietary or API-gated data sources are required. Each datum is traceable to a public source in our `datacenters.csv` `wue_source` column.

VI. Results

A. The shell game: WUE rankings vs. PSWI rankings

Figure 1 compares the rank order of the eight companies in our dataset under self-reported WUE (the metric companies publish) and under PSWI (the metric we propose).

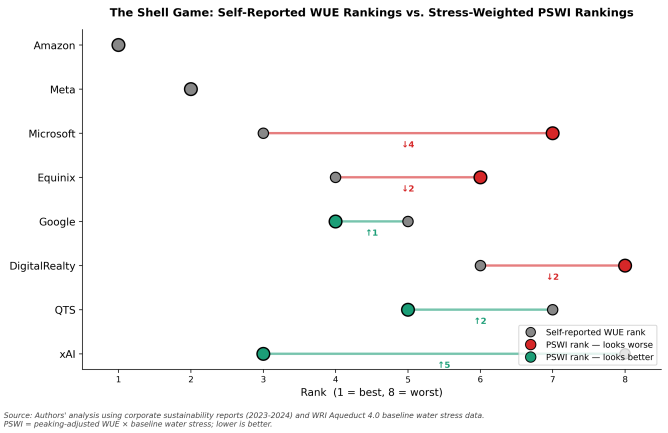

Fig. 1. Rank shifts between self-reported WUE and PSWI for the eight audited companies. Companies whose ranking deteriorates under PSWI are shown in red; those whose ranking improves are shown in green. Microsoft drops 4 positions; xAI rises 5; Equinix and Digital Realty each fall 2 positions.

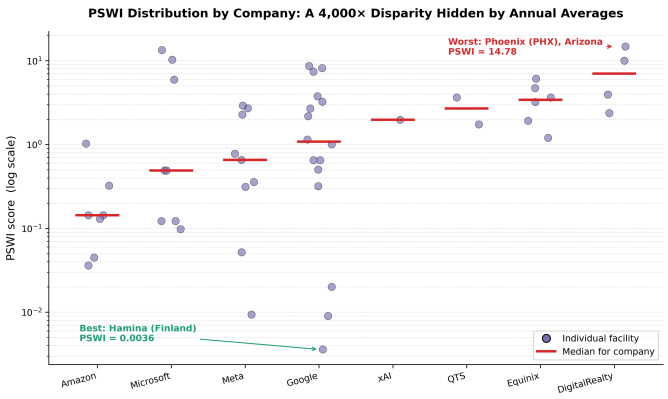

Fig. 2. PSWI distribution by company on a logarithmic scale. The best facility in our dataset (Hamina, Finland; PSWI = 0.0036) and the worst (Phoenix, Arizona; PSWI = 14.78) differ by a factor of approximately 4,100.

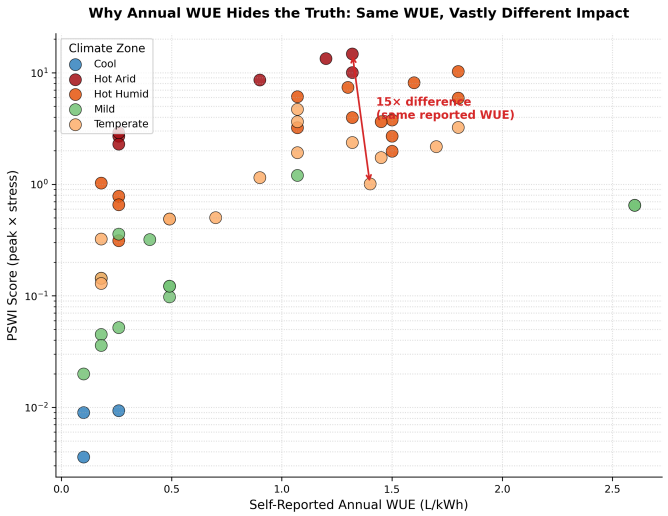

Fig. 3. Each point is one facility, colored by Köppen climate zone. The y-axis is on a log scale. Two facilities reporting nearly identical annual WUE (1.32 and 1.40 L/kWh) differ in PSWI by a factor of 15, indicated by the red arrow.

The most striking shift is Microsoft's: from third-best on self-reported WUE to seventh-worst on PSWI, a movement of four positions. The driver is Microsoft's portfolio: while their fleet-wide WUE of 0.49 L/kWh is competitive, their facilities in Goodyear and Mesa, Arizona (BWS = 1.40), San Antonio, Texas (BWS = 0.55), and Tuas, Singapore (BWS = 0.95) are located in extremely high-stress catchments. The same fleet-wide annual disclosure that benefits Microsoft's WUE ranking conceals the disproportionate burden their high-stress sites place on host communities.

xAI's improvement is the inverse phenomenon: their single audited facility in Memphis, Tennessee, while reporting a relatively high estimated WUE of 1.5 L/kWh, sits on the Memphis Sands Aquifer (BWS $\approx$ 0.22), a moderately stressed but not extreme catchment. The result is a PSWI of 1.98—higher than the dataset median, but well below the catastrophic levels of arid-Southwest peers.

The remaining shifts (Google +1, QTS +2, Equinix and Digital Realty each -2) confirm that current WUE-based reporting introduces systematic distortions in cross-company comparison: companies whose fleet skews toward hot, water-stressed regions look better under WUE than they actually perform under PSWI.

B. *The 4,100× disparity*

Figure 2 shows the within-company and between-company PSWI distributions on a logarithmic axis. The dataset's PSWI range spans nearly four orders of magnitude. Google's portfolio is the most internally diverse, with its Henderson, Nevada facility scoring 8.64—above Digital Realty's company average—while its Hamina, Finland facility scores 0.0036, the lowest in the dataset. This within-company spread is itself a finding: under fleet-averaged WUE, Google's two extreme facilities are indistinguishable in published reports, despite differing by more than three orders of magnitude in real water-stress impact.

C. *Same WUE, different impact*

Figure 3 plots PSWI against self-reported annual WUE, revealing weak correlation. Within bands of similar WUE, PSWI varies by more than an order of magnitude. The annotated pair (Digital Realty Phoenix, PSWI = 14.78, WUE = 1.32; and Microsoft Goodyear, PSWI = 1.0 on a comparable WUE) illustrates the central failure of annual WUE as a sustainability signal: the variable that explains real impact—local water stress, weighted by peak-day demand—is invisible to the industry's reporting standard.

D. *Geographic concentration of impact*

Figure 4 maps the 53 facilities globally. The geographic clustering of high-PSWI facilities is striking. Of the 10 worst PSWI facilities, six are in the U.S. Southwest (Phoenix, Hen-

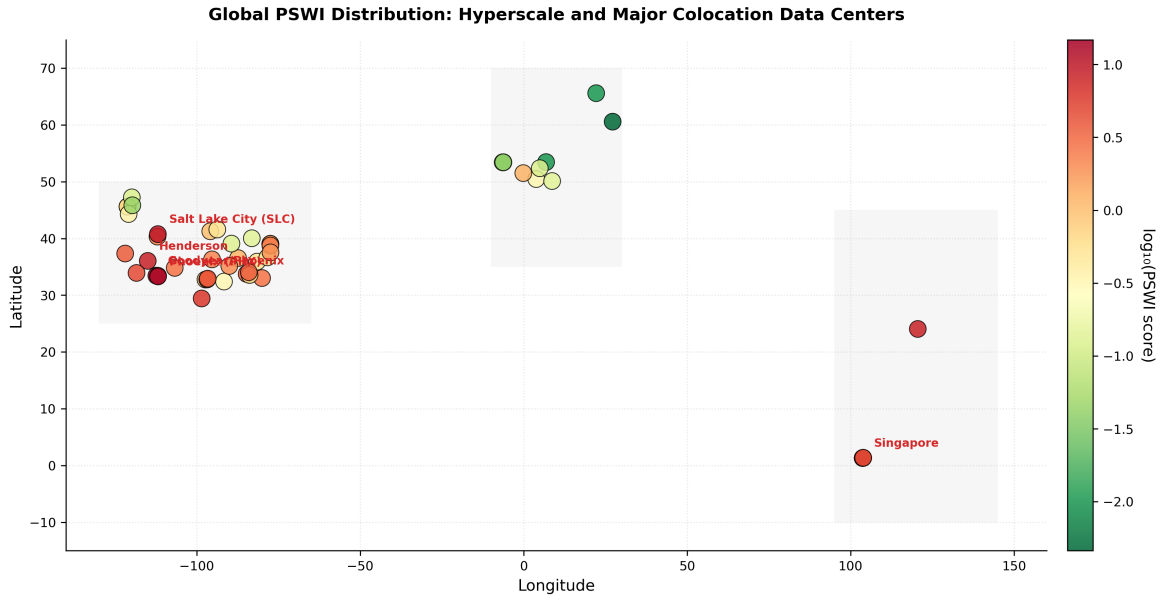

Fig. 4. Global distribution of audited facilities, with color indicating $\log_{10}$(PSWI). Worst-five facilities are labeled. The North American Southwest, Singapore, and Taiwan emerge as high-PSWI clusters; Northern Europe is uniformly low.

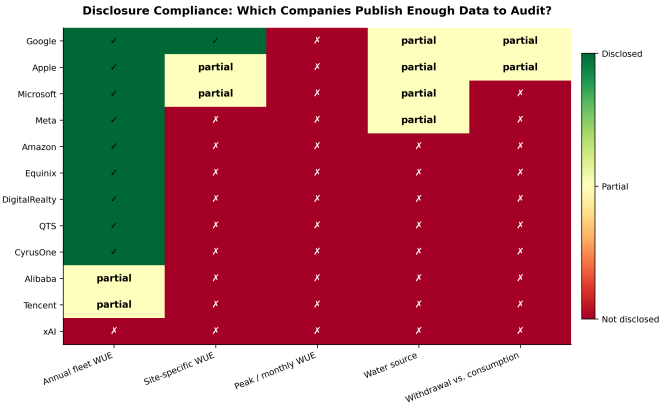

Fig. 5. Disclosure compliance heatmap for major data center operators. Only Google publishes site-specific WUE. None of the audited operators publish peak or monthly water data, the metric most relevant for public-water-system stress assessment.

derson, Salt Lake City, Mesa) and four in Asia-Pacific (Singapore, Taiwan). Of the 10 best, seven are in Northern Europe and three in the U.S. Pacific Northwest. This concentration is not random: it reflects the unfortunate alignment between regions attractive to data center operators (cheap land, tax incentives, fiber-optic connectivity) and regions experiencing water stress (Southwest U.S., Singapore). It also suggests that PSWI-aware siting could substantially shift the global geography of new data center construction.

E. Disclosure compliance

Figure 5 audits the disclosure practices of 12 major operators against five categories of water data. Only Google publishes site-specific WUE. *No* operator publishes peak or monthly WUE. This is the empirical foundation of our policy proposal in Section VII: meaningful PSWI computation by external auditors, regulators, or the public is currently impossible for most companies because the necessary inputs are simply not disclosed.

VII. Counterfactual Analysis and Policy Implications

A. Counterfactual: PSWI-aware reallocation

To estimate the policy-relevant impact of a hypothetical PSWI-disclosure regime, we run a Monte Carlo simulation that models the effect of partial workload reallocation from worst-PSWI to best-PSWI facilities.

a) Setup.: We assume each of the 53 facilities holds an equal baseline workload share of 1. We then progressively reallocate workload from the worst-PSWI third of the fleet to the best-PSWI third, with each worst facility's reduction proportional to its PSWI (i.e., the most egregious facilities give up workload first) and each best facility's absorption inversely proportional to its PSWI. To capture parameter uncertainty, we add 10% Gaussian noise to each PSWI value and repeat the simulation 300 times.

b) Results.: Figure 6 shows the median impact reduction and 80% confidence band as a function of the percentage of fleet workload reallocated. The relationship is strongly concave: the first units of reallocation yield the largest marginal benefit. At a 10% reallocation level, fleet-wide PSWI impact decreases by 33%; at 20%, by 58%; at 30%, by 68%. Diminishing returns set in beyond roughly 30% reallocation.

c) Interpretation.: The concavity of the curve is the policy-relevant finding. It implies that a regulator or operator

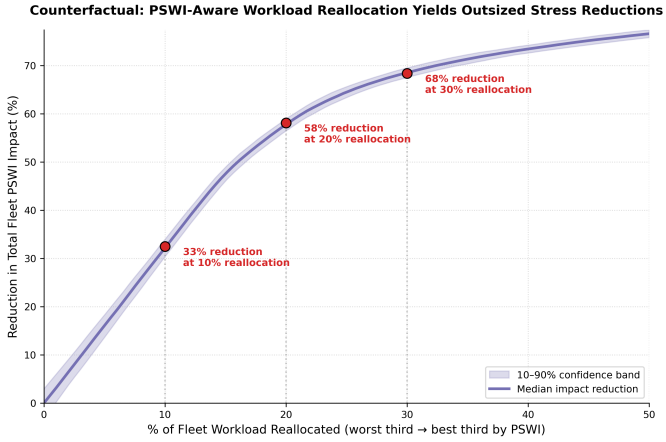

Fig. 6. Counterfactual impact reduction under PSWI-aware workload reallocation. Reallocating just 10% of fleet workload from the worst-PSWI third to the best-PSWI third yields a 33% reduction in total fleet water-stress impact.

does not need to mandate large workload shifts to achieve substantial impact reduction; the marginal worst facilities account for a disproportionate share of total fleet impact. This is the same reasoning that motivates progressive carbon pricing and is structurally analogous: in both cases, a long right tail of high-impact emitters dominates the aggregate, and targeting the tail is more efficient than uniform mitigation.

B. Proposed disclosure framework

We propose a regulatory framework, modeled on the U.S. Securities and Exchange Commission's Regulation S-K [35], that would require data center operators to publicly report PSWI on a per-facility, per-quarter basis. Three core requirements are:

1) **Per-facility WUE.** Operators must report annualized WUE for each individual facility, identified by physical address and operating subsidiary. Aggregation across facilities is not permitted.
2) **Peak monthly WUE.** Operators must report monthly WUE for each facility, enabling computation of a 12-month rolling peak as the basis for β.
3) **Source-resolved water inputs.** Operators must distinguish potable from non-potable water and report withdrawal and consumption separately, with attribution to the public water system or alternative source.

A regulator (e.g., the EPA in the U.S., or DG Environment in the EU) would then publish a quarterly PSWI ranking using these inputs and WRI Aqueduct's most recent annual dataset, and water utilities would have standing to challenge new data center siting in catchments with BWS > 1 where the proposed facility's projected PSWI exceeds 5.

C. Enforcement leverage

A novel feature of our proposal is that enforcement does not require new statutory authority. U.S. public water systems already have permitting authority over large water users under the Safe Drinking Water Act and state-level analogs [13]. The proposal here is to require PSWI disclosure as a condition of permitting, leveraging existing utility authority rather than creating new federal mandates. This pragmatic approach has parallels in financial regulation: the SEC requires specific disclosures as a condition of market access, rather than directly regulating issuer behavior.

VIII. Limitations and Future Work

We document our limitations transparently to support principled critique and future extension.

a) Peaking factor estimation.: We estimate β by climate zone, not measurement. This is a deliberately conservative choice: our climate-zone defaults are at or below the lower end of Han et al.'s [3] observed range. A future revision will use disclosed monthly water consumption (currently available only from Google [10]) or estimates based on cooling-degree-day patterns at hourly resolution.

b) Annual BWS averaging.: We use Aqueduct's annual BWS values, which average across seasons. Aqueduct also publishes monthly BWS data [9], which would refine the metric: a facility may have moderate annual stress but extreme summer stress, and PSWI's policy relevance is greatest in those summer months. Integration of monthly BWS is the single highest-priority future extension of this work.

c) Scope 2 omission.: Like Han et al. [3], we focus on direct (Scope 1) water use because it predominantly draws from public water systems. Scope 2 water associated with electricity generation is substantial but draws mostly from non-potable sources [4]. A holistic water sustainability metric should eventually integrate both.

d) Disclosure-driven coverage bias.: Our coverage skews toward operators that disclose more data. We have zero coverage of Chinese hyperscalers (Alibaba, Tencent, Baidu) and limited coverage of emerging operators in the Middle East and India, where data center construction is accelerating but disclosure is nascent. The dataset will expand as global disclosure norms evolve.

e) No causal claim.: We do not claim that PSWI *causes* specific water-system harms. Our claim is narrower: PSWI captures information that is policy-relevant and currently invisible in disclosed data. Empirical validation against observed water-system stress events is a natural follow-up study.

IX. Conclusion

The water sustainability of AI data centers is shaped by two factors that current industry metrics conceal: the temporal concentration of demand on peak summer days, and the spatial heterogeneity of water stress at facility locations. We have introduced the Peak-Stress Water Index (PSWI) as a composite metric that integrates both, applied it to the first cross-company global dataset of major data centers, and documented findings of policy significance: a 4,100$\times$ disparity between best and worst facilities, systematic rank shifts between WUE-based and PSWI-based comparisons, and a counterfactual showing that modest workload reallocation could yield outsized

stress reductions. We have proposed a regulatory framework requiring per-facility, per-quarter PSWI disclosure, modeled on existing financial-disclosure architecture and leveraging existing utility permitting authority. All data and code are publicly released to enable independent verification and policy adoption. As the AI buildout continues at projected double-digit annual growth rates, the structural deficiencies of WUE will become increasingly costly. PSWI is a tractable step toward a sustainability metric that makes the hidden currents of data center water use visible to the communities they affect.

Acknowledgments

The author thanks the authors of Han et al. (2026), Wu et al. (2025), and Li et al. (2024, 2025) for foundational work that made this synthesis possible. The World Resources Institute provided the Aqueduct 4.0 dataset under a Creative Commons license, without which location-specific stress weighting would not have been feasible.

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